Yan Zhuang

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Education

Rutgers University, Ph.D. in Economics — Sep. 2026 – Present

Renmin University of China, B.Econ. in Economics of Resources and Environment — Sep. 2022 – Jun. 2026

- GPA: 3.69/4.0

University of California, Davis, Global Study Program (Exchange) — Jan. 2025 – Mar. 2025

- GPA: 4.0/4.0

Research Interests

- Environmental Economics, Energy Economics, and Climate Change
- Applied Econometrics and Causal Inference

Research

Integrating Climate Change Impacts into the Assessment of China's Onshore Wind Power Potential — Jun. 2024 – Present

With Yanhua Wang. Under review at *Environmental Impact Assessment Review*

- Developed a multi-tiered (geographical, technical, economic) framework to project onshore wind power potential through 2100, using CMIP6 data (BCC-CSM2-MR, CAMS-CSM1-0, FGOALS-f3-L) under three SSP scenarios.
- Extended the levelized cost of energy (LCOE) model to incorporate the effect of temperature on turbine efficiency and of precipitation-induced blade erosion on operational lifetime.
- Revealed a mismatch between resource supply and demand and quantified regional economic risks: precipitation-related degradation raises LCOE by more than 10% in southern coastal provinces, offsetting gains from higher wind speeds.

Sand Out, Water In: The (Un)intended Consequences of Afforestation on the Hydrological Cycle and Agricultural Outcomes — Oct. 2024 – Present

Work in progress. Based on my undergraduate thesis, Renmin University of China

- Develop a quasi-experimental strategy that exploits high-frequency variation in prevailing wind direction to isolate atmospheric spillovers of afforestation on precipitation and agricultural productivity.
- Construct wind-mediated forest exposure measures from satellite forest cover, Lagrangian air-mass trajectories, and reanalysis wind data, separately identifying moisture advection and shelterbelt (wind-blocking) mechanisms.
- Provide causal evidence that upwind afforestation increases downwind wheat yields, even conditional on local precipitation and weather controls, consistent with non-precipitation atmospheric moisture channels.
- Quantify the aggregate economic value of these atmospheric externalities through a spatially explicit counterfactual analysis of China's Three-North Shelterbelt Program.

Research Projects

ABM-based Policy System Research on Express Packaging Waste Management Mar. 2024 – Mar. 2025

Project Leader. Awarded "Excellent Completion" distinction, Renmin University of China

- Led a team analyzing survey data from approximately 1,600 respondents (consumers, merchants, couriers) to parameterize the utility functions and interaction rules of heterogeneous agents.
- Built an agent-based model (ABM) to simulate dynamic behavioral responses to "Reduce, Replace, Recycle" policies, capturing dynamics that static equilibrium models cannot.
- Identified incentive-compatibility mechanisms: non-monetary incentives significantly outperform traditional subsidies in encouraging merchants to reduce packaging.

2024 Tsinghua International SDGs Case Analysis Competition Apr. 2024 – Aug. 2024

Honorable Mention, Group Leader

- Analyzed stakeholder relationships, incentive mechanisms, and practical effects under the Carbon Inclusive Mechanism framework.
- Proposed solutions for China under SDG 12: Responsible Consumption and Production.

Teaching

Teaching Assistant, Carbon Market Theory and Practice, Renmin University of China Fall 2025

- Selected by Dr. Wang to support undergraduate coursework based on academic excellence.

Honors and Scholarships

Academic Perfection, UC Davis Global Study Program Winter 2025

Academic Excellence Scholarship, Renmin University of China 2023, 2025

Professional Experience

Intern, Beijing Greenovation Institute for Public Welfare Development Aug. 2024 – Nov. 2024

- Monitored and summarized key proceedings of the 16th Conference of the Parties (COP16) to the UN Convention on Biological Diversity for internal briefings.
- Conducted comparative research on the establishment and cooperation of national parks in China and the U.S.
- Contributed research on China–ASEAN agricultural and food systems cooperation to a published policy report.
- Authored a research brief on the inclusion of gender perspectives in countries' Nationally Determined Contributions (NDCs).

Skills

Computing: R, Stata, Python, ArcGIS Pro, LaTeX

Languages: Mandarin Chinese (native), English (proficient)